

Design and Implementation of Low-Latency Sound Direction Estimation on Resource-Constrained Microcontrollers Using Dual-MEMS GCC-PHAT and TinyML

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Abstract—Background—Acoustic direction finding provides spatial awareness where cameras fail because smoke, darkness, occlusion, and privacy constraints do not suppress sound. **Problem—**Conventional arrays depend on multichannel converters, matrix-intensive estimators, graphics processors, or cloud execution, creating power, cost, latency, and confidentiality penalties for autonomous edge nodes. **Proposed Edge Pipeline—**This work presents a direction-of-arrival estimator on an ESP32-WROOM-32 and two INMP441 microelectromechanical-system microphones separated by 0.15 m. 16-kHz I2S samples are acquired through FreeRTOS direct-memory-access double buffers in 512-sample frames. Each channel undergoes mean removal, root-mean-square gating, and Hann tapering. A 512-point radix-2 fast Fourier transform forms the cross-power spectrum; phase-transform normalization suppresses coloration; an inverse transform yields a generalized cross-correlation whose maximum is searched over the admissible ± 7 -sample interval. Peak-to-sidelobe ratio rejects ambiguous frames, and the time difference is mapped to azimuth by a far-field propagation model with temperature-dependent sound speed. **Key Empirical Results—**Across 650 trials at 13 stations in a $6.2\text{ m} \times 4.5\text{ m} \times 3.0\text{ m}$ room, overall mean absolute error was 3.82° , with 1.40° broadside error and 91.8% of estimates within $\pm 5^\circ$. Measured digital-signal-processing time was 0.85 ms per frame, corresponding to less than 2.7% of a 240-MHz core over a 32-ms frame period. Average power was 138.9 mW. A TinyML roadmap replaces brittle peak selection under severe reverberation with a 514-feature one-dimensional convolutional regressor, quantized to INT8 for a 22.3-kB model and 18.5-kB tensor arena. The design demonstrates privacy-preserving spatial audition without cloud inference or accelerator-class hardware while retaining deterministic timing and an error model.

Index Terms—Direction of arrival (DoA), time difference of arrival (TDOA), generalized cross-correlation with phase transform (GCC-PHAT), MEMS microphones, I2S protocol, ESP32, TinyML, spatial audio.

I. INTRODUCTION AND MOTIVATION

SOUND is a complementary sensing modality for machines that must act when line of sight is unavailable. Acoustic waves propagate around moderate obstructions and remain observable in darkness, haze, and smoke; an optical detector can be blinded by illumination change or occlusion even while speech, an impact, or an alarm remains audible. Unlike imaging, a direction estimate need not preserve identity-bearing visual detail. Local processing can therefore expose only azimuth and confidence, reducing the privacy surface of occupancy, robotics, assistive interfaces, and safety-monitoring systems.

High-performance acoustic localization is typically demonstrated with four to dozens of synchronized channels, dedicated multichannel converters, desktop-class CPUs, digital signal processors, or GPUs. Such platforms may consume approximately 5–45 W depending on channel count and accelerator load, and their memory traffic, cooling, cost, and software stack are poorly matched to a battery-operated node. Cloud forwarding adds network-dependent delay and creates a raw-audio confidentiality risk. The design question is consequently not whether more sensors can improve localization, but what useful spatial estimate can be sustained under strict limits on energy, memory, channels, and response time.

A. Research objective

This work investigates the minimum practical architecture: one ESP32-WROOM-32 module and two clock-synchronous INMP441 microphones. The estimator obtains a one-dimensional azimuth in the half-plane convention -90° to $+90^\circ$. Its central hypothesis is that physically constrained GCC-PHAT, combined with explicit confidence gates, provides sufficient accuracy for coarse steering while leaving a compatible path to learned reverberation compensation.

B. Contributions and scope

The contributions are: (1) a complete geometric and spectral derivation tied to a 0.15-m baseline and 16-kHz sample rate; (2) a codec-free shared-I2S interface; (3) a bounded-memory DMA/FFT pipeline with a ± 7 -sample search; (4) an uncertainty policy based on dual-channel RMS and peak-to-sidelobe ratio (PSLR); (5) a Web Serial visualization that transports estimates rather than audio; and (6) an INT8 1D-CNN roadmap.

The system estimates one dominant far-field source. It does not claim full 360° identifiability, multi-source separation, elevation, or archival-grade validation from aggregate statistics alone. “Latency” is used precisely: 0.85 ms is processing time after a frame is available; the 512-sample observation interval is 32 ms and remains part of event-to-output delay.

Notation: microphone signals are $x_1[n]$ and $x_2[n]$; f_s is sample rate; d is baseline; c is sound speed; τ is delay; and θ is azimuth measured from broadside.

II. LITERATURE SURVEY AND RELATED WORK

A. Classical time-delay estimation

Knapp and Carter formalized generalized cross-correlation (GCC) as a family of frequency-weighted delay estimators [1]. For microphone spectra $X_1(f)$ and $X_2(f)$, the phase transform (PHAT) divides the cross spectrum by its magnitude. This equalizes spectral amplitudes, emphasizes interchannel phase consistency, and often sharpens the correlation peak for broadband signals. It does not, however, remove late reverberation: reflected paths create additional phase-consistent peaks, especially in short frames [2]. Robust TDOA variants exploit redundancy across more microphones [3], whereas the present design has only one pair and must instead constrain the search and reject low-confidence frames.

B. Subspace estimators and their cost

MUSIC decomposes the spatial covariance $R_{xx} \in \mathbb{C}^{M \times M}$ into signal and noise eigenspaces and scans a steering-vector pseudospectrum [4]. ESPRIT replaces the angular scan with an invariance relation between two matched subarrays [5]. Let a dense Hermitian eigensolver reduce R_{xx} by Householder transforms and diagonalize the result. Both stages require sums over three matrix indices, giving $\Theta(M^3)$ arithmetic; covariance storage is $\Theta(M^2)$. A grid search adds $\Theta(G M K)$ steering products for G directions and K frequency bins.

$$C_{\text{eig}}(M) = \alpha M^3 + O(M^2), \quad S_{\text{cov}}(M) = b M^2 \quad (1)$$

Here α depends on the eigensolver and b is bytes per complex element. Doubling array order asymptotically multiplies eigensolver work by eight and covariance storage by four. On a microcontroller this cost competes with DMA buffers, FFT workspaces, task stacks, and telemetry. For $M=2$, decomposition is numerically possible but offers little subspace separation and cannot resolve more than one source robustly. Thus the obstacle is not that a 2×2 eigendecomposition cannot run, but that the subspace method's resolution benefit requires the additional sensors, snapshots, calibration, and search work the target platform intentionally avoids.

C. Steered response and learned localization

SRP-PHAT evaluates phase-weighted pair correlations at hypothesized source locations [6]. For $P=M(M-1)/2$ pairs, G grid points, and K bins, direct frequency-domain steering costs $\Theta(PGK)$; lookup-based accumulation reduces constants but not the grid dependence. Coarse-to-fine search mitigates this overhead [7]. A single pair avoids spatial-grid summation and maps one lag directly to one principal azimuth.

Learning-based localization replaces explicit propagation inversion with a trainable mapping. SELDNet demonstrated that convolutional-recurrent networks can jointly detect events and regress directions from multichannel magnitude and phase features [8]. Its sequence model is too large for the baseline ESP32, but the principle motivates a compact 1D convolution over cross-spectral features. TensorFlow Lite Micro (TFLM) supplies static-memory inference for microcontrollers [9], while INT8 quantization can reduce weight storage by approximately fourfold when weights dominate graph overhead [10]. The roadmap in Section VI is therefore deliberately narrower than SELDNet: one source, one angle, one frame, and no recurrent state.

TABLE I
Comparative Analysis of Direction Estimators

Method	Compute / memory	Mics	MCU feasibility
XCORR	$\Theta(N \log N)$; $\Theta(N)$	2+	High; coloration-sensitive
GCC-PHAT	$\Theta(N \log N)$; $\Theta(N)$	2+	High; chosen baseline
SRP-PHAT	$\Theta(PGK)$; lookup tables	3+	Medium; grid-bound
MUSIC	$\Theta(M^3 + GMK)$; $\Theta(M^2)$	≥ 3 typical	Low at useful array order
ESPRIT	$\Theta(M^3)$; $\Theta(M^2)$	Matched subarrays	Low-medium; geometry-bound
SELDNet-class	Model-dependent MACs	4 typical	Low without compression
INT8 1D-CNN	Fixed quantized MACs	2	Prospective; TFLM

Definitions: N , frame length; M , microphones; P , microphone pairs; G , candidate directions; K , frequency bins; MAC, multiply-accumulate. Feasibility assumes bounded SRAM, no external accelerator, and real-time frame deadlines.

D. Design position

The surveyed methods establish a rational hierarchy. Raw cross-correlation is cheapest but vulnerable to spectral coloration. GCC-PHAT adds only per-bin normalization to the same FFT order and is robust enough to form a deterministic baseline. SRP-PHAT, MUSIC, and ESPRIT become attractive when more channels support actual spatial diversity. Learned localization is reserved for a second-stage correction after a reproducible analytical baseline exists. This separation also keeps failure analysis interpretable: each rejected frame can be traced to energy, peak structure, lag bounds, or geometry rather than an opaque posterior alone.

Complexity conclusion: The selected two-channel pipeline performs two forward FFTs and one inverse FFT per frame, i.e., approximately $\Theta(3N \log_2 N)$, plus $\Theta(N)$ normalization. No covariance eigendecomposition or direction grid is required.

III. THEORETICAL FOUNDATIONS AND MATHEMATICAL MODELING

A. Far-field propagation geometry

Place two microphones at positions $\pm d/2$ on the array axis and define $\theta=0^\circ$ as broadside. If source range $R \gg d$, spherical wavefront curvature over the aperture is negligible and the incident field is locally planar. Projecting the baseline onto the propagation direction gives the signed path-length difference

$$\Delta \ell = d \sin \theta, \quad -\pi/2 \leq \theta \leq \pi/2. \quad (2)$$

With sound speed c in metres per second, the interaural time difference is

$$\Delta t = \tau = \Delta \ell / c = (d/c) \sin \theta. \quad (3)$$

For $d=0.15$ m and $c=343.2$ m/s, the physical bound is $|\tau| \leq 0.00043706$ s ≈ 437.06 μ s. At $f_s=16,000$ samples/s, $|\tau f_s| \leq 6.993$ samples; the discrete search therefore uses lags $q \in \{-7, \dots, +7\}$. The principal-angle inverse is

$$\theta = \sin^{-1}(c \tau / d) \cdot 180^\circ / \pi. \quad (4)$$

The arcsine argument is clipped to $[-1, 1]$ only to prevent floating-point overflow; repeated clipping is logged as a fault because it indicates model, calibration, or peak-selection error. Subsample interpolation may fit a parabola around the selected correlation peak, but reported implementation timing should distinguish that optional step from integer-lag operation.

B. Temperature compensation

For dry air near ordinary indoor conditions, an ideal-gas approximation gives

$$c(T) = 331.3 \sqrt{1 + T/273.15}, \quad (5)$$

where T is air temperature in degrees Celsius and 273.15 converts to absolute temperature. Linearization about room temperature yields $dc/dT \approx 0.60$ m $\cdot$ s $^{-1}$ $\cdot$ °C $^{-1}$. Because (4) is proportional to c , uncompensated temperature creates a multiplicative sine error. At a fixed measured delay,

$$\partial \theta / \partial c = \tau [d \sqrt{1 - (c\tau/d)^2}]^{-1} \text{ rad} \cdot \text{s} \cdot \text{m}^{-1}. \quad (6)$$

Temperature metadata should therefore accompany each calibration run, particularly near endfire where the denominator approaches zero.

C. Spatial sampling

A two-sensor baseline is free of monochromatic spatial aliasing only when the maximum phase difference does not exceed π , giving

$$f_{\text{alias}} = c/(2d). \quad (7)$$

Using 343.2 m/s gives 1,144 Hz; the frequently rounded $c=343.0$ m/s gives 1,143.3 Hz. Speech and impulsive stimuli extend above this limit. PHAT broadband averaging does not repeal spatial sampling theory; it causes many frequency-dependent aliases to add less coherently than the true broadband delay. Reliable operation still benefits from band selection, a compact baseline, and confidence rejection.

D. Discrete GCC-PHAT derivation

For a frame of $N=512$ samples, first remove the sample mean μ_i and apply a Hann window:

$$y_i[n] = (x_i[n] - \mu_i) w[n], \quad w[n] = \frac{1}{2} - \frac{1}{2} \cos(2\pi n / (N-1)). \quad (8)$$

The discrete Fourier transforms are

$$X_i[k] = \sum_{n=0}^{N-1} y_i[n] e^{-j2\pi kn/N}. \quad (9)$$

The cross-power spectral density between channel 1 and channel 2 is

$$G_{12}[k] = X_1[k] X_2^*[k], \quad (10)$$

where $(\cdot)^*$ denotes complex conjugation. PHAT uses the weighting

$$\Psi_{\text{PHAT}}[k] = 1 / (|G_{12}[k]| + \epsilon), \quad (11)$$

with small $\epsilon > 0$ preventing division by zero. The normalized spectrum and generalized cross-correlation are

$$\hat{G}_{12}[k] = G_{12}[k] \Psi_{\text{PHAT}}[k], \quad (12)$$

$$r_{12}[q] = (1/N) \sum_{k=0}^{N-1} \hat{G}_{12}[k] e^{j2\pi kq/N}. \quad (13)$$

Under the ideal model $x_2(t) = a x_1(t - \tau_0)$ with positive gain a , (10) has phase $e^{j2\pi f \tau_0}$. Magnitude normalization removes $a|X_1|^2$, leaving a unit phasor. Its inverse Fourier transform approaches $\delta(\tau - \tau_0)$ for continuous bandwidth, or a periodic discrete impulse for the DFT. The estimator is

$$\hat{q} = \arg \max_{|q| \leq 7} r_{12}[q], \quad \hat{\tau} = \hat{q} / f_s. \quad (14)$$

If polarity or channel ordering can invert the peak, firmware must fix one sign convention by calibration; silently taking $|q|$ would destroy left/right information.

E. Statistical interpretation

PHAT approximates a maximum-likelihood delay cue when phase is more reliable than magnitude, but its whitening boosts weak noisy bins. The RMS gate suppresses frames without sufficient energy; PSLR checks whether a unique maximum exists. Neither gate is a probability. A defensible deployment calibrates thresholds against held-out noise and reverberation conditions and reports the rejection rate alongside angular error.

IV. HARDWARE ARCHITECTURE AND SENSOR INTEGRATION

A. Processing core

The ESP32-WROOM-32 integrates a dual-core 32-bit Xtensa LX6 processor clocked up to 240 MHz, 520 kB of on-chip SRAM, I2S peripherals, direct-memory access (DMA), and wireless connectivity [11]. The selected firmware uses FreeRTOS to isolate acquisition, processing, and telemetry. A hardware single-precision floating-point unit supports the baseline spectral arithmetic without an external DSP. Wireless radios are not required for estimation and may be disabled during power characterization; browser telemetry uses USB serial.

B. Digital acoustic front end

Each INMP441 combines an omnidirectional MEMS transducer, signal conditioning, a delta-sigma analog-to-digital converter, antialias filtering, and a 24-bit I2S output. The specified signal-to-noise ratio is 61 dBA [12]. Digital microphones eliminate interchannel codec routing and analog gain mismatch, but they do not eliminate timing, sensitivity, or package-placement errors. Both devices share serial clock (SCK), word select (WS), and serial data (SD). Their L/R pins select opposite time slots; the nonselected device tri-states SD, permitting a stereo bus.

TABLE II
Hardware Interconnect and Pin Assignment

Net / device pin	ESP32 / connection	Function
INMP441-L SCK	GPIO 26	I2S bit clock
INMP441-R SCK	GPIO 26	Shared bit clock
INMP441-L WS	GPIO 25	Left/right word clock
INMP441-R WS	GPIO 25	Shared word clock
INMP441-L SD	GPIO 33	Shared serial data
INMP441-R SD	GPIO 33	Tri-state multiplexed data
INMP441-L L/R	GND	Left time slot
INMP441-R L/R	3.3 V	Right time slot
Both VDD	3.3-V rail	Sensor supply
Both GND	Common ground	Signal reference

Each microphone should receive local supply decoupling close to VDD and GND. Trace length and return-path asymmetry should be minimized. GPIO selection is a firmware routing choice, not a fixed ESP32 electrical requirement.

C. Sampling and frame format

The I2S peripheral operates as master receiver at 16 kHz stereo. Although each microphone delivers 24 significant bits, a 32-bit slot is convenient for DMA alignment. One frame therefore contains 512 samples per channel, or 1,024 words. With 4-byte words, one interleaved frame occupies 4,096 bytes. Double buffering reserves 8,192 bytes before FFT and workspace storage.

$$T_{\text{frame}} = N/f_s = 512/16000 = 32 \text{ ms.} \quad (15)$$

The shared SCK and WS establish sample-level coherence. This point is essential: two independent I2S peripherals with unsynchronized clocks can accumulate fractional-sample drift and bias TDOA. The dual-slot bus instead expresses left and right samples within one I2S frame. Firmware converts sign-extended 24-bit words to normalized floating point or fixed-point values using the same scale for both channels.

D. Mechanical and acoustic constraints

Microphone acoustic ports lie on a rigid 0.15-m baseline. The delay model uses acoustic-center spacing, not PCB-edge spacing. A calibration fixture should measure port centers and verify channel order with a source near each end. Enclosures must expose both bottom ports through comparable cavities; unequal ports act as frequency-dependent filters and corrupt phase. To preserve the far-field model, the 1.50-m test range gives $R/d=10$. This is a pragmatic ratio, although residual curvature remains measurable near the aperture extremes.

E. Task partition and data ownership

The acquisition task receives DMA buffers and timestamps completed blocks. The DSP task owns a copied or ping-ponged frame while DMA fills the alternate buffer. A low-priority telemetry task formats only angle, quality, status, and RMS values. Raw audio is neither transmitted nor stored by the core design. A single-producer/single-consumer queue avoids locks inside the sample interrupt path; dropped frames increment a monotonic counter visible in telemetry.

Determinism conditions: (i) I2S DMA buffers must be in internal DMA-capable memory; (ii) FFT buffers must not alias a buffer still owned by DMA; (iii) serial writes must never block acquisition; and (iv) frequency changes and radio interrupts must be disabled or logged during timing trials.

F. Power measurement boundary

The reported 42.1-mA current at 3.3 V implies 138.9 mW at the measured rail. A reproducible report must state whether the USB-UART bridge, regulator loss, indicator LEDs, and radios are inside that boundary. Section IX treats the supplied figure as board-level project data and computes ideal battery duration without asserting regulator or battery conversion efficiency.

V. EMBEDDED FIRMWARE AND DSP PIPELINE

A. Streaming schedule

DMA descriptor A is processed while descriptor B is filled, then ownership is exchanged. The processing deadline is 32 ms. The critical path is: deinterleave and sign-extend; compute channel means and RMS values; subtract means and multiply by the Hann window; execute two forward FFTs; form and normalize the cross spectrum; execute one inverse FFT; unwrap circular lags; search ± 7 ; compute PSLR; and emit a validated estimate.

For each channel $i \in \{1, 2\}$, DC cancellation uses $\mu_i = N^{-1} \sum_n x_i[n]$. RMS is measured after mean removal:

$$\text{RMS}_i = \sqrt{[(1/N) \sum_{n=0}^{N-1} (x_i[n] - \mu_i)^2]}. \quad (16)$$

A frame is eligible only if both channels exceed a calibrated noise threshold γ_{RMS} . Dual gating prevents a disconnected or shadowed channel from producing an apparently sharp but meaningless phase estimate.

B. Radix-2 spectral engine

The 512-point in-place Cooley–Tukey FFT performs $\log_2 512 = 9$ butterfly stages. Bit-reversal reorders samples; at stage s , butterflies span 2^s samples. Precomputed twiddle factors avoid trigonometric calls in the real-time path. The inverse transform conjugates the twiddle sign and scales by $1/N$. One complex buffer may be reused after each channel spectrum has been retained, but lifetime analysis must ensure the cross spectrum is complete before either spectrum is overwritten.

C. Confidence policy

Let $p_0 = |r_{12}[\hat{q}]|$ be the selected peak. Exclude a guard band $B = \{q: |q - \hat{q}| \leq 1\}$ and define the largest admissible sidelobe $p_s = \max_{|q| \leq 7, q \notin B} |r_{12}[q]|$. Then

$$\text{PSLR} = p_0 / (p_s + \epsilon). \quad (17)$$

The frame is accepted when $\text{PSLR} \geq 1.6$ and both RMS gates pass. A ratio, rather than a decibel definition, is used here; 1.6 corresponds to $20 \log_{10}(1.6) = 4.08$ dB. Firmware and dashboard must label the ratio to prevent confusion with a dB PSLR.

D. Optimized GCC-PHAT kernel

```
// X1, X2: 512-point complex spectra; R reuses X1.
// fft512(a, inverse) is in-place radix-2.
constexpr int N = 512, QMAX = 7;
constexpr float EPS = 1.0e-12f;

for (int k = 0; k < N; ++k) {
    const float ar = X1[k].re, ai = X1[k].im;
    const float br = X2[k].re, bi = X2[k].im;
    // X1 * conj(X2)
    const float gr = ar*br + ai*bi;
    const float gi = ai*br - ar*bi;
    const float inv = 1.0f /
        sqrtf(gr*gr + gi*gi + EPS);
    R[k].re = gr * inv;
    R[k].im = gi * inv;
}
fft512(R, true); // IFFT includes 1/N

int bestQ = 0;
float peak = -1.0f;
for (int q = -QMAX; q <= QMAX; ++q) {
    const int idx = (q + N) & (N - 1);
    const float v = fabsf(R[idx].re);
    if (v > peak) { peak = v; bestQ = q; }
}

float side = 0.0f;
for (int q = -QMAX; q <= QMAX; ++q) {
    if (abs(q - bestQ) <= 1) continue;
    const int idx = (q + N) & (N - 1);
    side = fmaxf(side, fabsf(R[idx].re));
}
const float pslr = peak / (side + EPS);
const float tau = (float)bestQ / 16000.0f;
const float u = fminf(1.0f, fmaxf(-1.0f,
    soundSpeed * tau / 0.15f));
const float angleDeg = asinf(u) * 180.0f / M_PI;
```

The bit-mask index is valid because N is a power of two. A production implementation should retain an explicit status code for RMS rejection, PSLR rejection, clipping, and DMA overrun.

E. Numerical and timing safeguards

EPS regularizes silent bins but does not prevent them from receiving unit magnitude when noise dominates; band masks or coherence weights are candidate improvements. Single-precision arithmetic is adequate for unit-normalized spectra, provided input scaling avoids overflow before normalization. Cycle counts should be obtained with the CPU cycle counter around each stage, after cache warm-up, over many frames, and reported with median and high-percentile values. The supplied 0.85-ms figure is an aggregate compute measurement; stage-specific timings were not provided and are not synthesized here.

F. State machine

The estimator states are *ACQUIRE*, *PREPROCESS*, *CORRELATE*, *VALIDATE*, and *PUBLISH*. Any gate failure transitions to *PUBLISH* with an invalid status and last-valid angle held only for visualization. It must not be counted as a new success. This distinction prevents temporal smoothing from concealing estimator dropouts.

VI. TINYML ARCHITECTURE AND MACHINE LEARNING ROADMAP

A. Why learn after establishing DSP

In a reverberant room, the received channel is a convolution of the source with a room impulse response. The cross spectrum therefore contains direct-path and reflected-path phase terms; the largest GCC-PHAT peak need not represent the direct arrival. A compact regressor can learn recurring multipath structure, microphone mismatch, and sub-sample peak shape. The analytical pipeline remains valuable as a baseline, fallback, and source of physically meaningful features. This staged approach follows the broader evidence that learned phase/magnitude representations improve localization in noise and reverberation [8], [13].

B. A consistent 514-element feature

A real 512-point FFT has 257 unique bins from DC through Nyquist. To obtain the specified 514-element vector without double counting, discard DC, retain bins $k=1, \dots, 256$, concatenate the real and imaginary parts of the 256 PHAT-normalized cross-spectral bins (512 scalars), then append log-RMS from the left and right channels (2 scalars):

$$\mathbf{z} = [\text{Re}\{\hat{G}[1:256]\}, \text{Im}\{\hat{G}[1:256]\}, \log \text{RMS}_1, \log \text{RMS}_2] \in \mathbb{R}^{514}. \quad (18)$$

This convention includes Nyquist, whose imaginary component is ideally zero for real signals; replacing it with a coherence statistic is a possible refinement. The training and embedded feature extractors must implement identical bin order, scaling, epsilon, and window normalization. A saved feature schema should be versioned with the model.

C. Proposed one-dimensional CNN

The network accepts one 514×1 vector, applies a Conv1D layer with 16 filters, rectified linear activation, and max pooling, then flattens into Dense-64 and Dense-32 layers followed by one linear output θ_{ML} . Kernel width, stride, pool width, and padding must be selected during model development; without them, an exact parameter count or MAC count cannot be claimed. For kernel width h and pooled length L_p , the parameter count is

$$P = 16(h+1) + (16L_p+1)64 + (64+1)32 + 33. \quad (19)$$

A physically informed loss can combine angular absolute error with an endfire weight and a symmetry consistency term. Train/validation/test splits must be by recording session and room condition, not by randomly adjacent frames, to prevent leakage of nearly identical acoustic responses.

D. Quantization and deployment

Post-training integer quantization maps float activations r to integer q by $r \approx s(q - z_0)$, where $s > 0$ is scale and z_0 is zero-point. A representative calibration set must span angle, source type, level, temperature, and reverberation. Full integer operators then run through TFLM's static tensor arena [9]. Quantization-aware training is a fallback if post-training INT8 materially increases angular error [14].

The roadmap reports a 22.3-kB INT8 model, 74.8% smaller than Float32, implying an 88.5-kB Float32 artifact by $22.3/(1-0.748)$. The reported tensor arena is 18.5 kB. These are design measurements, not literature benchmarks. Latency and accuracy for either neural variant were not supplied and must be measured on the target ESP32 before comparative claims.

TABLE III
Footprint and Latency Comparison

Estimator	Flash / model	Peak SRAM	Latency
Float32 DNN	88.5 kB (implied)	Not reported	Not measured
INT8 1D-CNN	22.3 kB	18.5-kB arena	Not measured
GCC-PHAT	Not isolated	FFT/DMA buffers	0.85 ms measured

Interpretation: “74.8% reduction” applies to the serialized model, not necessarily total firmware flash. SRAM entries are not directly comparable until FFT workspaces, operator scratch buffers, stacks, and simultaneous buffer lifetimes are measured with one linker map and one runtime peak method.

E. Dataset and evaluation roadmap

- Acquisition:** store synchronized two-channel frames, geometry, room, temperature, source class, level, and ground-truth angle.
- Augmentation:** mix measured noise and room impulse responses without placing augmented siblings across data splits.
- Training:** optimize mean absolute angular error; monitor calibration by angle bins and source types.
- Conversion:** apply representative-dataset INT8 quantization; verify every operator is integer.
- On-device test:** measure arena high-water mark, latency distribution, energy per inference, and rejection-aware accuracy.

F. Decision fusion

A conservative deployment can run GCC-PHAT first and invoke the CNN only when PSLR lies in a gray zone. Agreement within a calibrated tolerance yields a valid estimate; disagreement sets an ambiguity flag. This cascade protects deterministic behavior while focusing learned computation on frames most affected by reverberation.

VII. REAL-TIME WEBGL 3D TELEMETRY DASHBOARD

A. Separation of estimation and presentation

The browser dashboard is an observer, not part of the estimator. React 19 manages connection and packet state, Three.js constructs the scene graph, and WebGL renders a spatial dome. The W3C Web Serial API establishes a driverless direct USB serial session at 115,200 bit/s after an explicit user gesture. Keeping DSP on the ESP32 ensures that display-frame jitter, browser throttling, or disconnection cannot alter the estimated angle.

B. Telemetry grammar

The firmware emits one newline-terminated ASCII record per accepted or rejected frame:

```
A:%.1f L:%.2f S:%d LR:%.0f,%.0f
```

Here **A** is angle in degrees, **L** is the linear PSLR ratio, **S** is a status integer, and **LR** contains left and right RMS values in the firmware's documented amplitude units. A receiver must buffer partial reads, split only at newline, reject malformed or nonfinite fields, range-check $-90^\circ \leq A \leq 90^\circ$, and treat unknown status codes as invalid. At 10 records/s and roughly 34 bytes/record, traffic is about 3.4 kbit/s including framing, far below 115.2 kbit/s.

C. Spatial scene

A titanium-colored baseline bar establishes the microphone axis; two golden MEMS packages mark acoustic centers at $\pm d/2$. A dynamic ray originates at the midpoint and rotates by the validated azimuth. Concentric wave rings expand from the source direction, while printed labels at $-90^\circ, -75^\circ, \dots, +90^\circ$ register the estimate against the laboratory grid. Color or opacity encodes validity and PSLR, not angle, so users can distinguish “no reliable estimate” from zero degrees.

Rendering uses a fixed physical convention: positive angle must correspond to the same microphone ordering used by (10)–(14). A one-time left/right calibration is displayed in the interface. Exponential animation smoothing is permitted for visual comfort, but the numeric readout shows the unsmoothed estimate and status. Otherwise the display could imply better temporal stability than the estimator delivers.

D. Serial and privacy considerations

The page requests a serial port only on user action and opens no network socket for core operation. It never receives PCM samples. The interface can log timestamped scalar packets for evaluation, but recording is opt-in and identifies dropped or malformed records. A strict parser limits line length and numeric ranges to prevent an attached device from allocating unbounded memory or injecting markup.

VIII. EXPERIMENTAL SETUP AND EVALUATION METHODOLOGY

A. Acoustic environment and geometry

Evaluation was specified in a $6.2 \text{ m} \times 4.5 \text{ m} \times 3.0 \text{ m}$ laboratory with reverberation time $RT60 \approx 0.42 \text{ s}$. $RT60$ denotes the time for sound energy level to decay by 60 dB after the source stops. The array center and acoustic ports were levelled; a laser-calibrated radial grid placed the source at $R=1.50 \text{ m}$. Thirteen stations covered -90° to $+90^\circ$ in 15° increments: $-90^\circ, -75^\circ, -60^\circ, -45^\circ, -30^\circ, -15^\circ, 0^\circ, +15^\circ, +30^\circ, +45^\circ, +60^\circ, +75^\circ$, and $+90^\circ$.

The specification reports 650 total trials using broadband claps/snaps, speech phonemes, frequency chirps, and pink noise. A complete archival protocol should state trials per angle and stimulus, source sound-pressure level, source height, microphone height, background level, temperature, and whether rejected frames remain in the denominator. Those raw allocation details were not supplied; the present analysis therefore does not manufacture them.

B. Ground truth and trial independence

Ground-truth angle θ_i is defined by the laser line from array midpoint to source position, using the same broadside sign convention as the firmware. Every trial should begin after moving or re-triggering the source so consecutive frames are not treated as independent when they share one acoustic event. Calibration trials must be excluded from test metrics. The 13 stations sample the field uniformly in angle, but not uniformly in delay because $\sin \theta$ is nonlinear.

C. Error metrics

For K valid trials, circular wrapping is unnecessary within $[-90^\circ, 90^\circ]$. Let $e_i = \hat{\theta}_i - \theta_i$. The prescribed metrics are

$$MAE = (1/K) \sum_{i=1}^K |e_i|, \quad (20)$$

$$RMSE = \sqrt{(1/K) \sum_{i=1}^K e_i^2}, \quad (21)$$

$$P_{\pm 5} = (100/K) \sum_{i=1}^K I(|e_i| \leq 5^\circ). \quad (22)$$

RMSE penalizes outliers more strongly than MAE. Because no per-trial errors or squared-error total were supplied, RMSE cannot be recovered from MAE alone and is reported as unavailable rather than inferred. The valid-frame rate K/N_{all} should accompany these metrics to expose accuracy–rejection tradeoffs.

VIII. EXPERIMENTAL SETUP AND EVALUATION METHODOLOGY (CONTINUED)

D. Stimulus rationale

Claps and finger snaps provide short broadband transients with steep onsets, offering strong cross-spectral phase support. Chirps provide controlled spectral occupancy and expose frequency-dependent phase inconsistencies. Pink noise distributes equal expected energy per octave and tests sustained broadband operation. Speech phonemes represent nonstationary, spectrally colored sources for the intended human-interaction domain. Results should be stratified by stimulus because one aggregate can conceal failure on narrowband or low-energy speech segments.

E. Sampling resolution and expected error structure

An integer lag step is $\Delta\tau=1/f_s=62.5\ \mu\text{s}$. At broadside, a one-sample change maps locally to

$$\Delta\theta \approx c/(df_s) = 0.1430\ \text{rad} = 8.19^\circ. \quad (23)$$

This first-order value appears coarser than the reported 1.40° broadside MAE. Such accuracy is possible only if the implementation obtains sub-sample information, aggregates over repeated frames, smooths outputs, or the reported statistic is based on a calibration mapping beyond a raw integer lag. The supplied code specification does not identify which mechanism produced the aggregate. Therefore an archival version must document interpolation or temporal aggregation explicitly. Reporting 0.1° telemetry precision does not itself imply 0.1° estimator resolution.

The FFT bin spacing is $\Delta f = f_s/N = 31.25\ \text{Hz}$. Frequency resolution governs the sampled spectrum, whereas zero-padding the cross spectrum can interpolate correlation time without adding information. A fair comparison must hold observation time, bandwidth, and post-processing constant.

F. Timing protocol

Compute time is measured from the first preprocessing instruction after DMA ownership transfer to the availability of angle and status, excluding the 32-ms acquisition period and optional serial blocking. For trial j , processor cycles ΔC_j convert to $t_j = \Delta C_j / 240\ \text{MHz}$. Report median, 95th percentile, maximum, compiler optimization, FFT implementation, flash/cache state, and task interference. The real-time utilization of one core is

$$U = t_{\text{DSP}} / T_{\text{frame}} \times 100\%. \quad (24)$$

G. Power protocol

Power is $P=VI$. Current should be measured with a calibrated inline instrument at the defined 3.3-V boundary over idle, acquisition, processing, and serial states. Average power requires integration over a complete operational cycle, not only a processor burst. If battery nominal capacity is Q_b ampere-hours and current is constant, ideal duration is

$$t_{\text{ideal}} = Q_b / I. \quad (25)$$

Real duration is lower after regulator efficiency, battery voltage limits, self-discharge, cell aging, radio activity, and capacity-rate effects. Energy per processed frame at the reported average power is $P T_{\text{frame}} = 4.44\ \text{mJ}$ if the complete board draws that power continuously; it is not the isolated DSP energy.

H. Reproducibility and statistical analysis

Per-trial records should include trial identifier, timestamp, true angle, estimate, validity status, PSLR, both RMS values, stimulus, temperature, and execution cycles. The primary analysis must predefine whether invalid frames count as failures for $P_{\pm 5}$. Confidence intervals can be obtained by bootstrap resampling at the independent acoustic-event level, not the frame level. A signed-error plot tests systematic channel or geometry bias; absolute error versus angle reveals endfire conditioning; error versus PSLR validates the confidence metric.

For comparisons among GCC-PHAT, an unweighted correlation baseline, and the future CNN, all algorithms should process the same held-out frames. Paired absolute-error differences enable a Wilcoxon signed-rank test when normality is doubtful. Report effect size and confidence interval rather than relying only on a p -value.

I. Threats to validity

- **Construct validity:** a stationary radial source tests azimuth estimation, not tracking latency.
- **Internal validity:** source placement, temperature, microphone mismatch, and repeated-event dependence can bias error.
- **External validity:** one room and one RT60 do not establish performance in corridors, vehicles, or outdoor wind.
- **Reporting validity:** aggregate MAE does not reveal tails, rejection rate, or per-stimulus behavior.

Minimum release package: raw scalar trial table, firmware commit identifier, build flags, microphone spacing measurement, room/temperature metadata, and scripts that recompute (20)–(25). Audio need not be released if privacy is a concern; feature vectors and labels can support audit.

IX. RESULTS, PERFORMANCE ANALYSIS AND BENCHMARKING

A. Accuracy

The project specification reports 650 trials, overall MAE of 3.82° , broadside MAE of 1.40° , and 91.8% of estimates within $\pm 5^\circ$. These three aggregates support a useful coarse-steering result, but they do not determine thirteen per-angle MAEs, RMSE, bias, variance, confidence intervals, or rejected-frame rate. Table IV therefore preserves every test station while marking unavailable cells. This avoids creating measurements that were not supplied.

TABLE IV
Empirical Accuracy Reporting Matrix Across 13 Angles

True angle	MAE	RMSE	Within $\pm 5^\circ$
-90°	Not reported	Not reported	Not reported
-75°	Not reported	Not reported	Not reported
-60°	Not reported	Not reported	Not reported
-45°	Not reported	Not reported	Not reported
-30°	Not reported	Not reported	Not reported
-15°	Not reported	Not reported	Not reported
0° (broadside)	1.40°	Not reported	Not reported
$+15^\circ$	Not reported	Not reported	Not reported
$+30^\circ$	Not reported	Not reported	Not reported
$+45^\circ$	Not reported	Not reported	Not reported
$+60^\circ$	Not reported	Not reported	Not reported
$+75^\circ$	Not reported	Not reported	Not reported
$+90^\circ$	Not reported	Not reported	Not reported
All stations	3.82°	Not derivable	91.8%

“Broadside” is interpreted as the reported 0° station because no broader angular subset was defined. If the project used a multi-station broadside region, Table IV must be updated from the raw trial file.

B. Execution latency

The reported total DSP time is 0.85 ms on a 240-MHz core, equivalent to approximately 204,000 cycles. Dividing by the 32-ms frame period gives 2.656%, consistent with the stated utilization below 2.7%. This leaves substantial deadline slack, but it is not a 0.85-ms event response: a newly arriving sound may wait up to one frame before processing begins.

TABLE V
Execution Latency on One 240-MHz Core

Stage	Time	Status
DMA/deinterleave	Not isolated	Included in total*
Mean/RMS/Hann	Not isolated	Included in total
Two 512-pt FFTs	Not isolated	Included in total
PHAT normalization	Not isolated	Included in total
512-pt IFFT	Not isolated	Included in total
Peak/PSLR/angle	Not isolated	Included in total
Total DSP	0.85 ms	Measured aggregate

*The exact start boundary was not supplied. The measurement should be repeated with the definition in Section VIII-F. Stage values are intentionally not apportioned without instrumentation.

C. Power and endurance

At 3.3 V and 42.1 mA, measured power is

$$P = 3.3 \times 0.0421 = 0.13893 \text{ W} = 138.9 \text{ mW}. \quad (26)$$

For a 1,000-mAh battery under the ideal constant-current assumption,

$$t_{\text{ideal}} = 1000 \text{ mAh} / 42.1 \text{ mA} = 23.75 \text{ h}. \quad (27)$$

Thus “>23.7 h” is supported only as ideal arithmetic at the stated current. A real LiPo system requires a regulator and has voltage-dependent usable capacity; measured endurance may be lower. The calculation also assumes that dashboard transport does not activate a higher-power radio state.

D. Benchmark interpretation

The strongest result is the ratio between compute time and frame period: the analytical estimator consumes about one thirty-eighth of a core deadline. The accuracy summary is encouraging but incomplete at endfire, where geometry predicts larger error. The TinyML footprint is plausible within module resources, yet no on-device latency or accuracy has been reported. Accordingly, GCC-PHAT is the demonstrated baseline and the CNN remains a roadmap, not a measured replacement.

Claim boundary: Measured/project-reported: 3.82° overall MAE, 1.40° broadside MAE, 91.8% within $\pm 5^\circ$, 0.85-ms aggregate DSP, 42.1 mA at 3.3 V. Derived: 2.656% utilization, 204,000 cycles, 138.9 mW, and 23.75-h ideal endurance.

X. DISCUSSION, LIMITATIONS AND FAILURE MODES

A. Cones of confusion and front-back ambiguity

A linear two-microphone array measures only the projection of the propagation vector onto its baseline. All directions satisfying the same projection produce the same TDOA; in three dimensions they form a cone about the baseline. Under the paper's two-dimensional half-plane convention, θ and $180^\circ - \theta$ share $\sin \theta$, so front and back cannot be distinguished. The displayed -90° to $+90^\circ$ angle is therefore a principal solution selected by an assumed source half-plane, not a unique 360° bearing.

Amplitude differences cannot reliably resolve this ambiguity for nominally omnidirectional matched microphones, because enclosure scattering and source level are confounded. A noncollinear third sensor, directional baffle, or multimodal cue is required for identifiable front/back classification.

B. Geometric dilution of precision at endfire

Let the measured discrete delay be q samples. Substituting $\tau = q/f_s$ into (4) in radians gives

$$\theta(q) = \sin^{-1}[cq/(df_s)]. \quad (28)$$

Differentiation yields the requested sensitivity:

$$d\theta/dq = c/[df_s \sqrt{1 - (cq/df_s)^2}] = c/(df_s \cos \theta). \quad (29)$$

Here θ is the principal angle and the derivative is radians per sample. If σ_q is small lag uncertainty, first-order propagation gives $\sigma_\theta \approx |d\theta/dq| \sigma_q$. At broadside, $|\cos \theta| = 1$ and sensitivity is minimized. As $|\theta| \rightarrow 90^\circ$, $\cos \theta \rightarrow 0$ and uncertainty diverges. This is the geometric reason an overall aggregate must be accompanied by per-angle error; no signal-processing method can remove the conditioning of the two-sensor inverse map.

C. Temperature and baseline error

Using $u = ct/d$, a first-order perturbation is

$$\delta\theta \approx [u/\sqrt{1-u^2}](\delta c/c - \delta d/d). \quad (30)$$

With $dc/dT \approx 0.6$ m/s per $^\circ\text{C}$, a 10°C error changes sound speed by about 1.75%. The angular effect is modest near broadside and amplified near endfire by the same $1/\cos \theta$ term. Baseline measurement error is equally multiplicative with opposite sign. A low-cost temperature sensor and measured acoustic-center spacing are therefore more valuable than hard-coding 343.2 m/s.

D. Reverberation and correlated interference

PHAT flattens magnitude but cannot identify which coherent path is direct. Early reflections can produce a stronger admissible peak than the line-of-sight path; late diffuse reverberation broadens the peak and lowers PSLR. A fan, loudspeaker, or vibration mechanically coupled to both sensors can create coherent interference and pass an energy gate. Remedies include frequency masking, coherence weighting, adaptive noise models, temporal tracking with explicit dropout states, and the learned residual estimator proposed in Section VI.

E. Spatial aliasing and source bandwidth

The 0.15-m aperture aliases monochromatic components above roughly 1.144 kHz at endfire. Broadband impulses may still produce a unique group-delay peak because the correct delay aligns phase slopes across frequency. Narrowband whistles and tonal machinery remain difficult: several lags can fit nearly identical phase observations. PSLR should reject many such cases, but rejection rate must be reported. Reducing d raises the alias limit while worsening time-delay resolution, revealing a fundamental aperture tradeoff.

F. Synchronization, clipping, and mismatch

- **Clock/frame fault:** swapped I2S slots reverses azimuth; bit-slip or independent clocks bias phase.
- **Clipping:** a saturated impulse creates harmonics and a misleadingly sharp correlation. Peak sample magnitude should be monitored.
- **Port mismatch:** unequal cavities introduce group-delay error not removed by DC cancellation.
- **Gain mismatch:** PHAT reduces amplitude sensitivity, but RMS gating and quantization still depend on calibrated channel scales.
- **DMA overrun:** a missed descriptor can pair noncontemporaneous blocks. Sequence numbers must invalidate the frame.

G. Latency semantics

The measured 0.85 ms is excellent computational latency, yet causal response includes acoustic propagation, frame accumulation, scheduling, processing, serialization, and rendering. With nonoverlapping 512-sample frames, acquisition contributes 0–32 ms depending on event phase, plus 0.85 ms compute. Overlapped windows can reduce update interval but not the information requirement of the window. Claims should use “0.85-ms DSP execution” rather than “0.85-ms end-to-end latency.”

H. Measurement completeness

The reported 1.40° broadside MAE is substantially smaller than the 8.19° one-sample local spacing from (23). This does not make the result impossible, but it requires documented sub-sample estimation, aggregation, or calibration. Likewise, 91.8% within $\pm 5^\circ$ cannot reveal performance if a large fraction of frames were rejected before scoring. These are not cosmetic reporting issues: they determine whether another laboratory can reproduce the claim.

I. Deployment implications

The design is most credible as a low-power directional trigger for camera steering, robot orientation, or human-machine interaction, with an explicit invalid state. It is not a safety-certified locator, a multi-talker tracker, or a substitute for a calibrated array. Processing scalar telemetry locally reduces unnecessary data exposure, but any saved timestamps or direction histories may still reveal occupancy patterns and should follow retention limits.

XI. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper specified and critically analyzed a two-microphone direction-of-arrival system for a resource-constrained ESP32. The architecture acquires coherent 24-bit I2S channels at 16 kHz, processes 512-sample frames with mean removal and Hann tapering, computes GCC-PHAT through two forward FFTs and one inverse FFT, limits the peak search to the physically valid ± 7 samples, and rejects uncertain estimates using dual RMS gates and $\text{PSLR} \geq 1.6$. Geometry maps accepted delay to principal azimuth with temperature-dependent sound speed.

The supplied study aggregates report 3.82° overall MAE, 1.40° broadside MAE, 91.8% success within $\pm 5^\circ$, 0.85-ms DSP execution, and 138.9-mW board power. Derived utilization is 2.656% of one 240-MHz core per 32-ms frame; ideal 1,000-mAh endurance is 23.75 h at the stated current. These figures support low-cost, private edge direction sensing. They do not establish end-to-end 0.85-ms response, per-angle robustness, or neural-model superiority. Raw station-level errors, rejection counts, RMSE, stage timings, and model inference measurements remain necessary for archival verification.

B. Future work

A four-microphone planar array can remove the single-baseline front/back ambiguity and estimate two-dimensional azimuth robustly; a tetrahedral geometry can extend tracking to elevation. Multiple baselines also enable SRP-PHAT or robust redundant TDOA fusion. ESP32-S3 vector instructions should be evaluated for FFT, INT8 convolution, and fused activation kernels. The proposed 514-feature 1D-CNN requires leakage-safe training, representative INT8 calibration, on-device latency profiling, and matched-room testing. Temperature sensing, fractional-delay interpolation, calibrated acoustic ports, overlapped frames, and a published trial-level dataset are immediate priorities. Standard DSP texts [17] and array references [18], [19], together with reproducible TinyML benchmark practice [15], [16], provide the methodological basis.

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Symbols and constants

Symbol	Definition	Value / unit
d	Acoustic baseline	0.15 m
c	Sound speed at test model	343.2 m/s
f_s	Sampling frequency	16 kHz
N	Frame / transform length	512 samples
q	Discrete delay	samples
τ	Continuous TDOA	seconds
θ	Broadside azimuth	degrees/radians
M	Microphone count	2
K	Valid scored trials	count
ϵ	Numerical regularizer	positive scalar

Reproducibility note: The requested aggregate results are reproduced exactly. Values not present in the project brief—per-angle errors, RMSE, frame rejection rate, neural inference latency, and stage timing—are explicitly marked unavailable rather than estimated. This manuscript is suitable for defense review; those records should be inserted before external archival submission.